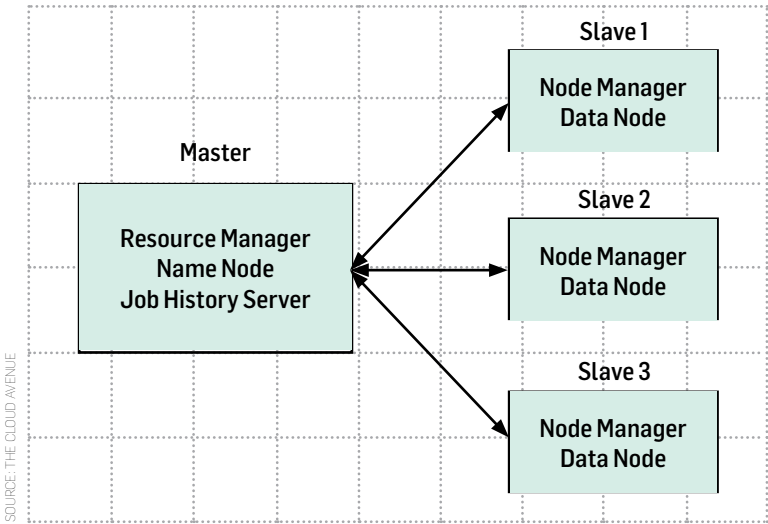




The Master-Slave node structure in Hadoop

HDFS is based on the master/slave architecture where master consists of a single NameNode that manages the file system metadata and single/multiple slave(s) DataNodes that store the actual data.

A file in an HDFS namespace gets split into many blocks and these blocks are stored in a set of DataNodes. The NameNode determines the mapping of these blocks to the different DataNodes. The DataNodes takes care of read and write operation with the file system. They are also responsible for taking care of block replication, creation and deletion based on instruction given by NameNode.

♦ **Hadoop MapReduce:** Hadoop MapReduce is a software based framework for writing applications that process large amounts of structured, semi-structured and unstructured data stored in the HDFS.

The term MapReduce can be splitted into two different categories of tasks :

- (a) **The Map Task:** This is the first task, which takes input data and converts it into a set of data and then individual elements are broken down into a finite ordered list of elements or tuples (key/value pairs).
- (b) **The Reduce Task:** This task takes the output from a map task as input and combines those data arranged as a finite ordered list of elements into a smaller set of list of elements. The reduce task is always performed after the map task.